

Paper I External Audit — Environment & Reproducibility

Independent Environment Record — PAPER_I_EXT_AUDIT

Audit ID: PAPER_I_EXT_AUDIT_2026-07-10

Date: 2026-07-10

Auditor: external AI adversarial auditor (adversarial-math-audit protocol)

Target (only): Paper I — *Affine Profile Reduction for Fractional Triangle Packings

in Split Graphs*: $|E(G)| - 2 \cdot v^*(G) \leq n^2/6 + n$ for split graphs G on n vertices.

Ground truth of the statement: 01_manuscript/PAPER_I_preprint_v1.0.{md,pdf,tex}, Theorem 1.1 (and Appendix C for the Lean statement).

Canonical package audited: ../github-sync/#81/official/preprints/PAPER_I/.

Independence statement

All auditor code was written from scratch for this audit (scripts/01-03, _lib/).
The numerical/symbolic checks recompute v^* , the residual-cover LP, the orbit LP, and every algebraic identity independently of the manuscript and its ledger. The Lean Gate-H build (scripts/04) is run in a **fresh copy** of the release sources placed in a clean directory with **no inherited** .lake **build artifacts**, using the community Mathlib olean cache — not the author's build cache. The author's repo of record (C:\Users\jtrav\aristotle_lean) was **not** used to produce any result here.

Software environment

Component	Version
OS	Windows 11 Home Single Language 10.0.26200
Python	3.14.4
sympy	1.14.0 (exact symbolic algebra — Script 01)
scipy	1.17.1 (HiGHS LP — Scripts 02, 03)
numpy	2.4.4
reportlab	5.0.0 (PDF rendering; no LaTeX/pdflatex available)
Lean toolchain	leanprover/lean4:v4.28.0 (pinned by lean-toolchain)
Mathlib	v4.28.0 (pinned rev per lake-manifest.json)
elan / lake	elan 4.2.3 ; lake 5.0.0 (Lean 4.28.0)

Master seed: 20260710 (fixed in Scripts 02, 03).

Objects (as computed)

- $v^*(G)$ — fractional triangle packing: $\max \sum_T w_T$ s.t. per-edge load ≤ 1 , $w \geq 0$.

- Residual cover $LP\ M(\kappa) = \min_{\{z \in P_p\}} \sum_e (1 - \kappa_e) z_e$, $P_p = \{z \in [0,1]^{E(K)} : z(T) \geq 1 \ \forall \text{ clique triangle } T\}$.
- $R(p, q) = (2p^2 - 2pq - q^2)/12$; A, B, C, U, D, H as in §6.

Gate-H build note (Windows long paths)

Mathlib contains file paths exceeding the Windows 260-char limit; with `LongPathsEnabled = 0` and `core.longpaths` unset, a checkout under a deep path fails (`git exit 128, "Filename too long"`). The independent build therefore uses (i) a short build root `C:\Users\jtrav\l`, and (ii) `core.longpaths=true` passed to git via `GIT_CONFIG_*` environment variables (no modification of the user's global git config). This is an environment workaround only; it does not affect the Lean sources or result.

Reproducibility

Each `scripts/NN_*/` folder holds the script, `results.txt` (full output), an English `report.md/.pdf`, a `.zip` of the folder, and its `.sha256`. A top-level `SHA256_MANIFEST.txt` lists digests of all files.